

Project	Strip footing on layered soil	Client	Tester evaluation
Prepared by	GeoEQ Cap tester build	Checked by	_____
Date	2026-09-05 22:17	Software	GeoEQ Cap 1.0.0 · Tester version · engine b1c4fd6a58bd
Company	_____	Model file	Strip footing on layered soil

Ultimate gross contact pressure 393.3 – 418.3 kPa

Numerical bracket width 6.18%

Interval midpoint (informative) 405.8 kPa



The model is numerically verified and contained for the associated-flow idealisation, but entered dilation is below friction for Sand. Run the Davis alternative or a documented sensitivity check before design use.

Model	Strip B = 2 m, T = 0.5 m, Df = 1 m, rough base
Soil profile	Upper sand over Stiff clay
Mesh	2,449 elements · 1 pass(es)
Cross-check	Indicative comparison — layered idealisation: method of characteristics 851.2 kPa (103.5% above)

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GeoEQ Cap
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1 Model definition

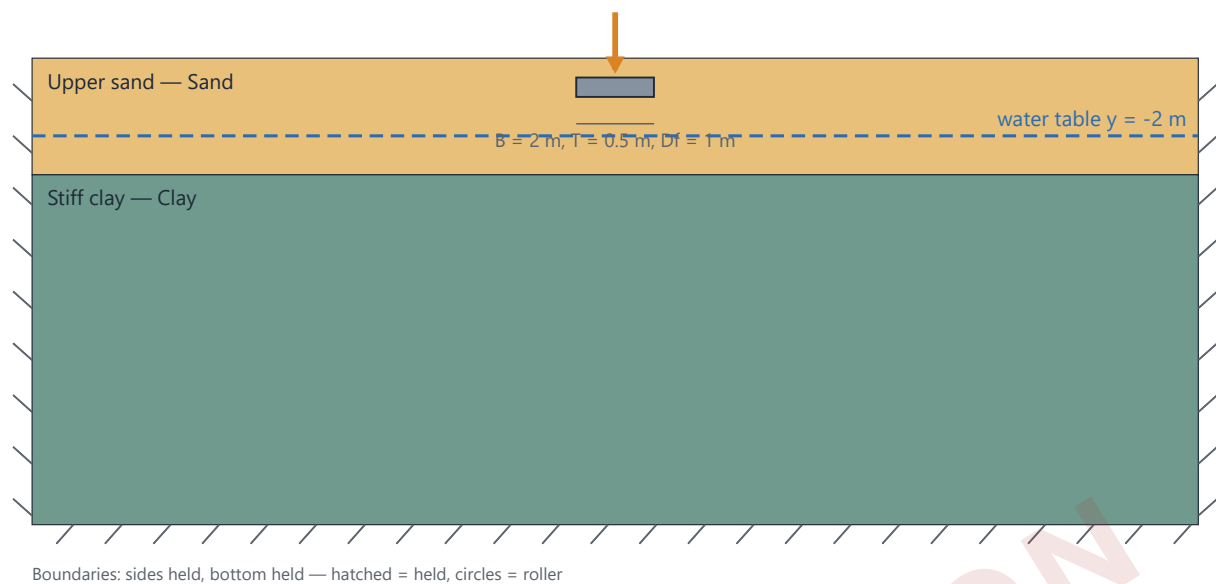


Figure 1 — Model as analysed: layers, water, loading and truncation supports (hatched = held, circles = roller). Elevations are absolute, as on the borehole log.

1.1 Geometry and boundaries

Footing	Strip $B = 2\text{ m}$, $T = 0.5\text{ m}$, $D_f = 1\text{ m}$, rough base, rigid
Analysis domain	$x = -15 \dots 15\text{ m}$, $y = -12 \dots 0\text{ m}$ · half model about $x = 0$
Truncation boundaries	auto — sides held, bottom held (drawn in Figure 1)
Water	water table at $y = -2\text{ m}$

1.2 Soil profile and material parameters

Layer	From – to (m)	Drainage	c' / s_u (kPa)	φ' (°)	ψ (°)	γ (kN/m ³)	γ_{sat} (kN/m ³)
Upper sand	0 → -3	drained	0	32	0	18	20
Stiff clay	-3 → -12	undrained	40	—	—	17	19

Table 1 — Characteristic soil parameters as entered.

1.3 Loading

Objective	ultimate bearing resistance
Footing load	vertical, central
Pressure definition	structural contact pressure at the footing base; footing self-weight is a separate fixed load
Seismic	none (static)

2 Method and assumptions

The bearing resistance is computed by **finite-element limit analysis (FELA)**: a lower bound from a statically admissible stress field (equilibrium, traction continuity and the Mohr–Coulomb criterion satisfied everywhere), and an upper bound from a kinematically admissible collapse mechanism (velocity boundary conditions and the associated flow rule). For a certified plane-strain case, the collapse load of the stated idealisation lies between the two bounds. The interval width is a numerical bracket, not a statistical confidence interval, allowable pressure or substitute for design factors. Both fields are re-verified after solving by independent admissibility checks (§4).

2.1 Analysis settings

Formulation	Plane strain — strip footing
Yield criterion	Mohr–Coulomb (exact second-order cone; no linearisation)
Design framework	Analysis only — nominal ultimate resistance
Edition	Code-neutral
Jurisdiction / annex	not specified
Design method	Unfactored characteristic resistance
Factor basis	Characteristic soil, actions and resistance are unfactored.
Dilatancy	Associated flow ($\psi = \varphi$)
Solver	Clarabel interior-point SOCP · QDLDL

Design scope: For research/benchmarking. The ultimate resistance is not an allowable pressure or a complete code design.

2.2 Assumptions made on your behalf

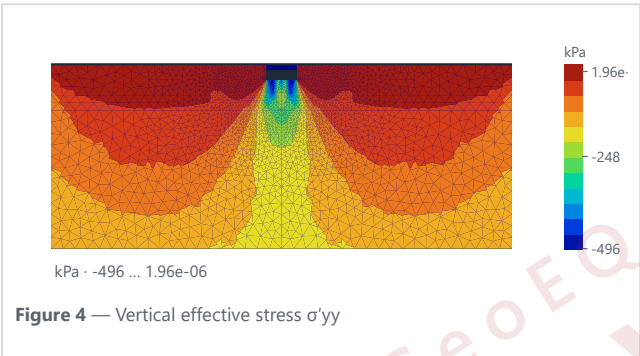
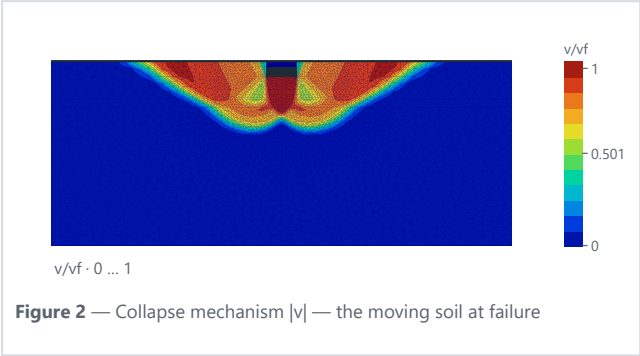
- A1** The footing is perfectly rigid; the load case is a direction in V–H–M space.
- A2** Plane-strain conditions; results are per metre run of an infinitely long strip.
- A3** Soil layers are homogeneous with the parameters of Table 1; strength gradients, where entered, vary linearly with depth.
- A4** Associated flow ($\psi = \varphi$). Associated flow is assumed although $\psi < \varphi$ was entered for: Sand — the computed resistance may be non-conservative. The Davis equivalent-strength option is the conservative alternative.
- A5** The footing is treated as weightless.
- A6** Truncation boundaries: sides held, bottom held. Domain adequacy is verified in §4.2.
- A7** Static conditions; no seismic inertia.
- A8** Hydrostatic pore pressure below the water table at $y = -2$ m; buoyant unit weights in the body force and effective-stress yield throughout.

Assumptions are restated with their consequences in the solver record (Appendix A); reviewers can reference them by number.

3 Results

Ultimate gross contact pressure	393.3 – 418.3 kPa
Numerical bracket width	6.18%
Interval midpoint (informative)	405.8 kPa

Table 2 — Result summary. 2,449 elements · 1 pass(es).



All figures are computed solver fields. Each colour scale records the displayed numerical range; robust ranges saturate the top and bottom 1% so isolated singularities cannot flatten an intensity plot.

4 Verification and quality assurance

GeoEQ Cap never trusts the optimiser's opinion of itself. After each solve, the returned fields are checked against every admissibility requirement by independent code; the bound status derives from these checks, not from the solver's termination report. Numerical admissibility and engineering model quality are reported separately: a passing field check does not override a failed domain-containment or modelling-assumption check.

4.1 Admissibility checks

Check	Requirement	Measured	Limit	Status
Lower bound — stress field				
Equilibrium	satisfied in every element	5.6e-15	1e-06	✓
Traction continuity	across every interior edge	3.4e-16	1e-06	✓
Boundary conditions	pointwise on every tagged edge	2.2e-17	1e-06	✓
Yield criterion	no strength deficit anywhere	8.9e-07	1e-04	✓
Upper bound — mixed (quadratic) formulation				
Weak equilibrium	every virtual-work row	8.7e-16	1e-06	✓
Yield criterion	at every quadrature point	0.0e+00	1e-04	✓
Duality gap	primal and dual agree	1.6e-05	1e-02	✓

Residuals are relative to the field scale; the yield deficit is in units of cohesion against the local strength.

4.2 Domain adequacy

Boundary	Near-boundary soil speed	Threshold	Status
Sides (far field)	0.0% of footing speed	5%	✓ contained
Bottom (rigid base)	0.0% of footing speed	5%	✓ contained

The failure mechanism does not press against the truncation boundaries, so the bracket holds for the real ground, not only the modelled domain.

4.3 Solver status

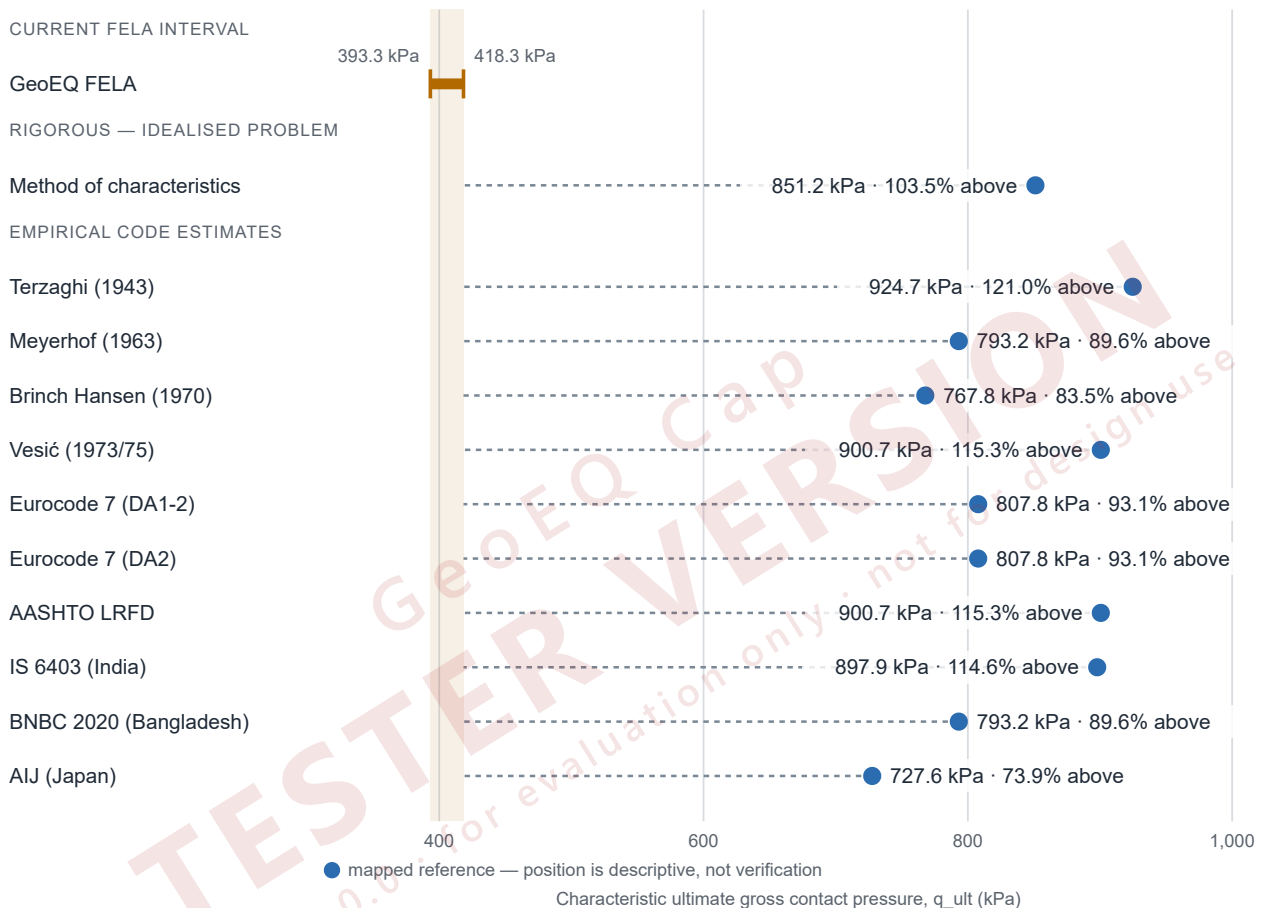
lower bound: NumericalError · upper bound: AlmostSolved — interior-point termination short of the tightest tolerance is normal for limit analysis, where the optimal field is not unique. These checks establish numerical admissibility for the stated model; the cover verdict also includes domain and assumption checks. Full detail appears in Appendix A.

5 Independent capacity comparison

Indicative comparison — layered idealisation — current analysis requires action

GeoEQ Cap calculates the modelled soil profile by finite-element limit analysis; its interval is the anchor of this comparison. The method of characteristics is a rigorous slip-line solution of the stated homogeneous idealisation. Conventional rows apply published correction-factor procedures. Agreement is a cross-check; disagreement identifies discretization or modelling-assumption sensitivity.

- The FELA model retains every layer. Reference methods extend the footing-base material 'Sand' as a homogeneous half-space; this is an indicative sensitivity only, not verification of the layered model.
- Embedment is represented by effective surcharge at footing-base level.
- Water effects use SlipBear effective surcharge and unit-weight mapping.



Characteristic ultimate gross contact pressure, kPa. Reference geometry: strip; reference soil: Sand. The FELA interval includes a documented footing-weight normalization of 0.000 kPa. Percentages outside the interval are measured from the nearest bound (above the upper bound or below the lower bound); they are not differences from the interval midpoint.

Ultimate comparison

Method	Result class	$q_{ult,gross}$	vs FELA interval	Interpretation
GeoEQ Cap FELA	Certified Interval	393.3–418.3 kPa	Anchor	Admissible interval for the current model box; the engineering quality check requires action before design use.
Method of characteristics	Rigorous (idealised problem)	851.2 kPa	103.5% above	Homogeneous extension of the footing-base soil; indicative only, not verification of the layered model.
Terzaghi (1943)	Empirical code estimate	924.7 kPa	121.0% above	Superposition of three separate mechanisms plus Terzaghi's own (approximate) factor charts; no depth contribution.
Meyerhof (1963)	Empirical code estimate	793.2 kPa	89.6% above	Superposition; $N_y = (N_q - 1)\tan(1.4\phi)$ sits below the exact rough-strip value at low ϕ and above it at high ϕ .
Brinch Hansen (1970)	Empirical code estimate	767.8 kPa	83.5% above	Superposition; $N_y = 1.5(N_q - 1)\tan\phi$ is a deliberately safe lower estimate (close to exact smooth-strip, below exact rough-strip).
Vesic (1973/75)	Empirical code estimate	900.7 kPa	115.3% above	Superposition; $N_y = 2(N_q + 1)\tan\phi$ generally overshoots the exact rough-strip value (unconservative at high ϕ).
Eurocode 7 (DA1-2)	Empirical code estimate	807.8 kPa	93.1% above	Superposition with $N_y = 2(N_q - 1)\tan\phi$ (rough base) and empirical shape factors; no depth factors.
Eurocode 7 (DA2)	Empirical code estimate	807.8 kPa	93.1% above	Superposition with $N_y = 2(N_q - 1)\tan\phi$ (rough base) and empirical shape factors; no depth factors.
AASHTO LRFD	Empirical code estimate	900.7 kPa	115.3% above	Vesic superposition formula with $N_y = 2(N_q + 1)\tan\phi$; LRFD factor applies to the gross nominal resistance.
IS 6403 (India)	Empirical code estimate	897.9 kPa	114.6% above	Net-pressure superposition with Vesic N_y and the W' water reduction; shape/depth factors are code-empirical.
BNBC 2020 (Bangladesh)	Empirical code estimate	793.2 kPa	89.6% above	Meyerhof-form superposition with global FoS; same theoretical limitations as the classical methods.
AIJ (Japan)	Empirical code estimate	727.6 kPa	73.9% above	Meyerhof-based superposition with AIJ α/β shape coefficients (β absorbs the $1/2$ factor); fixed FoS 3.

Design and allowable values

Method	Design/allowable value	Safety format	Basis note
Terzaghi (1943)	320.2 kPa	FoS = 3 (on net)	Informative conventional value; formats are not ranked.
Meyerhof (1963)	276.4 kPa	FoS = 3 (on net)	Informative conventional value; formats are not ranked.
Brinch Hansen (1970)	317.9 kPa	FoS = 2.5 (on net)	Informative conventional value; formats are not ranked.
Vesic (1973/75)	371.1 kPa	FoS = 2.5 (on net)	Informative conventional value; formats are not ranked.
Eurocode 7 (DA1-2)	389.9 kPa	partial factors (DA1-2)	Informative conventional value; formats are not ranked.
Eurocode 7 (DA2)	582.2 kPa	partial factors (DA2)	Informative conventional value; formats are not ranked.
AASHTO LRFD	450.3 kPa	LRFD $\phi_R = 0.5$	Informative conventional value; formats are not ranked.
IS 6403 (India)	370.0 kPa	FoS = 2.5 (net)	Informative conventional value; formats are not ranked.
BNBC 2020 (Bangladesh)	328.1 kPa	FoS = 2.5 (net)	Informative conventional value; formats are not ranked.
AIJ (Japan)	242.5 kPa	long-term $q_a = q_u / 3$	Informative conventional value; formats are not ranked.

Method of characteristics after Prandtl (1920) and Sokolovskii (1965); implementation validated against Martin's ABC (2003, 2004). Reference engine b1c4fd6a58bd. Comparison data are the immutable snapshot created with this FELA result.

6 References

- Sloan, S.W. (1988). Lower bound limit analysis using finite elements and linear programming.
- Sloan, S.W. & Kleeman, P.W. (1995). Upper bound limit analysis using discontinuous velocity fields.
- Makrodimopoulos, A. & Martin, C.M. (2006, 2007). Lower and upper bound limit analysis of cohesive-frictional materials using second-order cone programming.
- Martin, C.M. (2003, 2005). ABC — Analysis of Bearing Capacity; exact bearing capacity by the method of characteristics.
- Prandtl, L. (1920) · Sokolovskii, V.V. (1965). Statics of granular media.
- EN 1997-1:2004+A1:2013 Eurocode 7 (first generation).
- AASHTO (2024). LRFD Bridge Design Specifications, 10th edition.
- International Code Council (2024). International Building Code, Chapter 18; ASCE/SEI 7-22 load combinations.
- IS 6403 · BNBC 2020 · AIJ recommendations.

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A Appendix — full solver record

Method, every assumption made on your behalf, the mesh, the verification evidence for each bound, and the solver detail — exactly as the analysis ran. Nothing in the body of this report exists only in the summary.

GeoEQ Cap – bearing capacity by finite element limit analysis

Project : Strip footing on layered soil

Company : -

Date : 2026-09-05 22:17

Model : Plane strain

Geometry

Footing type strip, rigid
Footing Width B 2 m
Thickness T 0.5 m
Embedment Df 1 m
Base interface rough
Contour x -15 to 15 m, y -12 to 0 m
Symmetry half model about x = 0
Boundaries automatic (sides held, bottom held)
Water table y = -2 m

Loading

Analysis objective ultimate bearing resistance
Footing load vertical, central
Pressure definition structural contact pressure at the footing
 base; footing self-weight is a separate fixed
 load; no automatic gross/net conversion
Seismic none (static)

Soil profile

Layer From To Material
Upper sand 0.00 -3.00 Sand
Stiff clay -3.00 -12.00 Clay

Materials

Sand drained c' = 0 kPa, phi' = 32 deg, psi = 0 deg
 gamma = 18, gamma_sat = 20 kN/m3
Clay undrained su = 40 kPa
 gamma = 17, gamma_sat = 19 kN/m3

Method and assumptions

Computed : both bounds.
UB element : both upper-bound elements were run and the tighter valid bound kept (the smaller of two upper bounds is itself an upper bound).
Design basis: Analysis only – nominal ultimate resistance.
Design basis: characteristic values; no code check applied.
Edition : Code-neutral.
Jurisdiction: not specified.
Method : Unfactored characteristic resistance
Factor basis: Characteristic soil, actions and resistance are unfactored.
Scope : For research/benchmarking. The ultimate resistance is not an allowable pressure or a complete code design.
Lower bound : statically admissible stress field, linear within each element and discontinuous across element edges.
Upper bound : kinematically admissible mechanism with velocity discontinuities on every interior edge.
Yield : Mohr-Coulomb in exact second-order cone form. No linearisation of the yield surface.
Flow : ASSOCIATED (dilation angle = friction angle), which limit analysis requires. A real soil dilates less.
The upper bound survives that: an upper bound computed with $\psi = \phi$ is still an upper bound when $\psi < \phi$. The lower bound does NOT – proving it needs the exact solution to obey the flow rule. So for a dilatant material the result below is rigorous for the associated idealisation, and against the real soil only the upper bound is guaranteed.

WARNING: dilation is below friction in Sand.
The capacity reported here can exceed the real

one substantially – tens of percent for dense sand. Re-run with the Davis flow rule before relying on this number.

Stress : effective. Below the water table the body force is the buoyant weight $\gamma' = \gamma_{\text{sat}} - 9.81 \text{ kN/m}^3$.
Footing side: smooth rigid contact, compression only. Side friction is ignored, which is conservative.
Truncation : the far field and rigid base are artificial boundaries. The bounds are rigorous for the modelled body; check that the mechanism does not reach them.

Mesh

Coarseness medium (5 elements per half-width)
Edge refinement 4x at the footing edge
Elements 2,449
Nodes 1,287
Quality (gamma) min 0.61, mean 0.94
Elements under footing 9

Result

Lower bound 393.26 kPa
Upper bound 418.35 kPa
Interval midpoint 405.80 kPa (informative; half-width 3.09%)

The true collapse load is guaranteed to lie between the two bounds. This is a bracket, not an estimate.

Total load / metre run 786.53 to 836.69 kN/m
Solved loaded length 1 m (half model; total load doubled above)
Stopped: single mesh

Independent capacity comparison

Status Indicative comparison – layered idealisation
Mapping note The FELA model retains every layer. Reference methods extend the footing-base material 'Sand' as a homogeneous half-space; this is an indicative sensitivity only, not verification of the layered model.
Mapping note Embedment is represented by effective surcharge at footing-base level.
Mapping note Water effects use SlipBear effective surcharge and unit-weight mapping.
Pressure basis Characteristic ultimate gross contact pressure, kPa
FELA result class certified interval
Reference geometry strip
Reference soil Sand
FELA gross interval 393.26 to 418.35 kPa
Footing-weight addition 0.000 kPa

method	q _{ult} gross	vs FELA interval
Method of characteristics	851.17 kPa	103.5% above
Terzaghi (1943)	924.74 kPa	121.0% above
Meyerhof (1963)	793.22 kPa	89.6% above
Brinch Hansen (1970)	767.77 kPa	83.5% above
Vesic (1973/75)	900.66 kPa	115.3% above
Eurocode 7 (DA1-2)	807.83 kPa	93.1% above
Eurocode 7 (DA2)	807.83 kPa	93.1% above
AASHTO LRFD	900.66 kPa	115.3% above
IS 6403 (India)	897.88 kPa	114.6% above
BNBC 2020 (Bangladesh)	793.22 kPa	89.6% above
AIJ (Japan)	727.59 kPa	73.9% above

GeoEQ FELA is the numerical comparison anchor for the current model box.
The method of characteristics is a rigorous slip-line solution of the stated homogeneous idealisation; conventional rows are formula-based estimates, not equivalent numerical bounds.
Reference engine b1c4fd6a58bd

Verification

Lower bound – the returned stress field was tested directly:
equilibrium residual 5.6e-15
traction continuity 3.4e-16
boundary conditions 2.2e-17
yield violation 8.9e-07
verdict statically admissible
Upper bound – the returned field was tested directly:
weak equilibrium 8.7e-16

yield violation	0.0e+00
duality gap	1.6e-05
verdict	admissible and optimal

Optimality is checked as well as feasibility, and for this element that is not pedantry. It is the OPTIMUM of the mixed problem that bounds the collapse load from above; a merely feasible point sits below it, on the wrong side. The value reported is taken from the dual objective, which bounds the optimum from above at every iteration and therefore cannot be unsafe even if the solve is stopped early.

These checks are what make the numbers defensible. The optimiser's own termination status is not relied upon: limit analysis problems are degenerate and it often stalls with the objective already converged (lower: NumericalError, upper: AlmostSolved).

Solver

Lower bound 22,042 unknowns, 7,347 cones, 0.98 s
Upper bound 22,042 unknowns, 7,347 cones, 2.07 s
Clarabel interior-point SOCP, QDLDL linear solver.

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